

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think>

- 1 . The question asks about the feedback loop in men regulating sex hormone production .
 - 2 . To understand the feedback loop , we need to consider the key components involved in male sex hormone production .
 - 3 . The hypothalamus produces gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to release luteinizing hormone (LH) and follicle-stimulating hormone (FSH).
 - 4 . LH and FSH then stimulate the production of sex hormones by the testes in men (in women it's the ovaries).
 - 5 . Testosterone , being a sex hormone , has a suppressive effect on the production of GnRH from the hypothalamus and LH from the pituitary gland , thus completing the feedback loop .
 - 6 . Now , let's evaluate the options provided :
 - A , E , and F are involved in the feedback loop .
 - G is a sex hormone itself , not a regulator in the feedback loop .
 - B , the oxytocin , is related to social bonding and milk production during childbirth , not sex hormone regulation .
 - D , the hypothalamus , plays a crucial role by initiating the loop with GnRH .
- Therefore , the answer is the hormone involved in but not directly involved